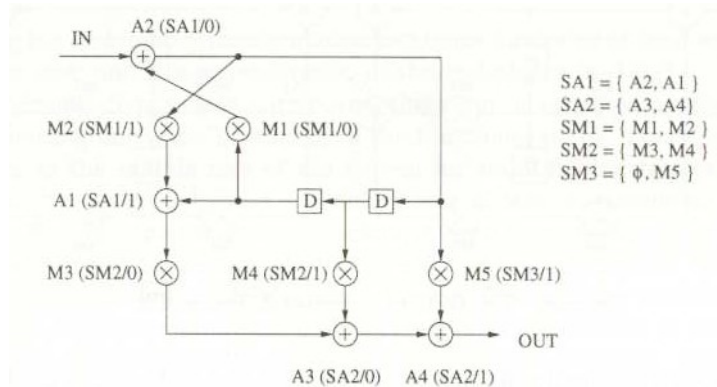


2019 Spring VLSI DSP Homework Assignment #4

For the design shown below.



Part A. (due date: 2019/6/4)

- For a folded design with 5 folding sets (SA1, SA2, SM1, SM2, SM3) and a folding factor equal to 2, verify if this is a valid folding
- If it is not, find a new valid folding design by modifying the folding orders in each set.
Note: Each folding set has only two possible folding orders. In total, there are 32 possible combinations of folding orders. You may write a program to enumerate all 32 possible combinations and check if any of them are valid.
- In case, you cannot identify any valid combinations of folding orders in (b), perform retiming first followed by folding
- Derive your folded design without register minimization based on the folding obtained in (b) or (c). You need to draw the folded circuit design.

Part B. (due date 2019/6/11)

- Write Verilog designs for both the original and the folded designs. Use Verilog simulation results to verify your folded design is correct. Assume inputs are 8-bit wide signed integers. Multiplier coefficients are all 8-bit wide and have magnitudes between 0.5 and 1. You may determine the word length of other variables so that no overflow occurs.
- Repeat (d) and (e) using the forward backward register allocation scheme to minimize the register usage. **Note:** you need to show the derivation details, e.g. life time chart, register allocation table, and so on.